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AWS Sustainable Infrastructure

Designing, building, and operating more efficient data centers

Sustainable infrastructure for every AI workload

AWS is building data centers to support the next generation of artificial intelligence (AI) innovation and customers' evolving needs. These capabilities combine innovations in power, cooling, and hardware design to create a more energy efficient data center that will underpin further customer innovation.



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The future of AWS data center design



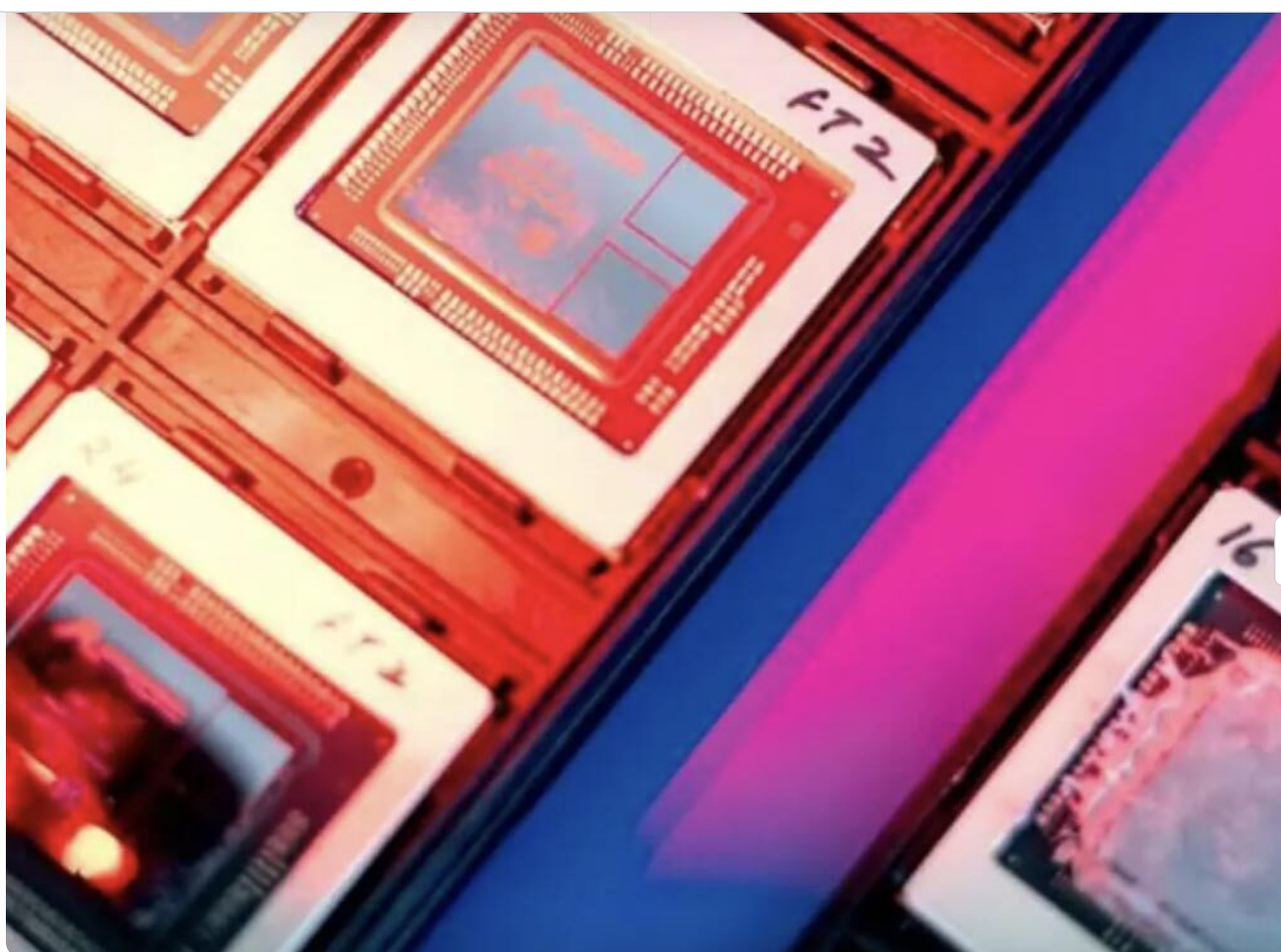
New data center components to support AI innovation

In 2024, AWS reported a global power usage effectiveness (PUE) of 1.15. PUE is one way we measure the efficiency of our data center operations. A lower PUE indicates a more efficient data center and a PUE score of 1.0 is perfect. AWS' latest innovative data center design delivers 12% more compute power with improved availability and efficiency.



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Accelerate innovation with AWS Silicon Chips

Customers can accelerate their sustainability innovation using AWS Silicon chips and leverage them to run their workloads. A recent study shows that customers running workloads on AWS silicon achieve energy efficiency by up to 99%, with reductions in carbon footprint by up to 81% using AWS's purpose-built silicon.



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AWS-designed silicon for efficient workloads

AWS Graviton

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cloud workloads running on Amazon EC2. Over 70,000 customers have used AWS Graviton to build efficient and performant workloads.

Optimizing efficiency across our operations



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Liquid cooling

Advanced rack design

Power usage effectiveness

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Flexible cooling design for AI processors

AWS has developed novel mechanical cooling solutions providing configurable liquid-to-chip cooling in both new and existing data centers. Some AWS technologies utilize network and storage infrastructure that does not require liquid cooling, so updated cooling systems will seamlessly integrate air and liquid cooling capabilities for the most powerful AI chipsets, like AWS Trainium3 and rack-scale AI supercomputing solutions like NVIDIA GB200 NVL72, as well as AWS's network switches and storage servers. This flexible, cooling design allows us to provide maximum performance and efficiency at the lowest cost, whether running traditional workloads or AI models.

Circular econon resources



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of spare parts, sourced from AWS own reuse inventory in 2024. And we expect this number to grow as we expand our efforts

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components recycled or sold on the secondary market

of plastic server rack components made of recycled or bio-based plastic

extended lifespan of Amazon hard disk drives



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Becoming water positive at AWS

In 2022, we announced our commitment to being water positive by 2030. That means we'll return more water to communities and the environment than we use in our data center operations. By the end of 2024, AWS was 53% of the way towards water positive.

There are four pillars of our water positive commitment:

- Water efficiency
- Sustainable sources
- Water reuse in communi
- Water replenishment



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